

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA0612	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	<i>Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)</i>		
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Section / Batch:	B TECH CSBS	Date:	27.07.2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I SHAFFRIA.A (192571047) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				
Clarity	10%				

Total Marks (100):

Faculty In-charge	Course Coordinator	Head of Department
Name: _____	Name: _____	Name: _____
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q75. Closest Pair – Hospital Location Planning

A city map provides hospital coordinates as $\{(1,2), (3,5), (6,7), (2,3), (8,1), (5,6)\}$. Apply the Closest Pair Brute Force algorithm to identify the nearest pair of hospitals. Clearly interpret the problem and justify the use of the brute force approach. Compute the distances between all possible pairs and show the calculations clearly. Identify the closest pair and count the total number of comparisons performed. Analyze the time complexity of the method. Verify the correctness of the result and interpret its significance in healthcare access planning.

Q76. Convex Hull – Boundary Detection in Urban Layout

Apply the Brute Force Convex Hull algorithm on the set $\{(0,0), (2,5), (5,7), (8,4), (7,0), (4,2), (3,3)\}$. Clearly interpret the problem and explain the concept of identifying the convex boundary. Check all possible line pairs and

determine which edges belong to the hull. Show the reasoning used to distinguish boundary points from interior points. Count the total comparisons performed. Analyze the time complexity and interpret the practical significance of the result in urban mapping.

(15).

CLOSEST PAIR.

PROBLEM

INTERPRETATION & JUSTIFICATION OF BRUTE

FORCE APPROACH.

→ We are given a set of Hospital coordinates in city map. $P = \{(1,2), (3,5), (6,7), (2,3), (8,1), (5,6)\}$. Our goal is to find pair of hospitals that have minimum euclidean distance between them.

While more efficient algorithms like divide and Conquer exist ($O(n \log n)$), the brute force method is straightforward, easy to implement and ideal for very small datasets ($n=6$) where overhead of complex algorithms is unnecessary.

$$\rightarrow d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\checkmark d(P_1, P_2) = \sqrt{(3-1)^2 + (5-2)^2} \approx 3.61$$

$$\checkmark d(P_1, P_3) = \sqrt{(6-1)^2 + (7-2)^2} \approx 7.07$$

$$\checkmark d(P_1, P_4) = \sqrt{(2-1)^2 + (3-2)^2} \approx 1.41$$

$$\checkmark d(P_1, P_5) = \sqrt{(8-1)^2 + (1-2)^2} \approx 7.07$$

$$\checkmark d(P_1, P_6) = \sqrt{(5-1)^2 + (6-2)^2} \approx 5.66$$

$$\checkmark d(P_2, P_3) = \sqrt{(6-3)^2 + (7-5)^2} \approx 3.61$$

$$\checkmark d(P_2, P_4) = \sqrt{(2-3)^2 + (3-5)^2} \approx 2.24$$

$$\checkmark d(P_2, P_5) = \sqrt{(8-3)^2 + (1-5)^2} \approx 6.40$$

$$\checkmark d(P_2, P_6) = \sqrt{(5-3)^2 + (6-5)^2} \approx 2.24.$$

$$\checkmark d(P_3, P_4) = \sqrt{(2-6)^2 + (3-7)^2} \approx 5.66.$$

$$\checkmark d(P_3, P_5) = \sqrt{(8-6)^2 + (1-7)^2} \approx 6.32.$$

$$\checkmark d(P_3, P_6) = \sqrt{(5-6)^2 + (6-7)^2} \approx 1.41.$$

$$\checkmark d(P_4, P_5) = \sqrt{(8-2)^2 + (1-3)^2} \approx 6.32.$$

$$\checkmark d(P_4, P_6) = \sqrt{(5-2)^2 + (6-3)^2} \approx 4.24.$$

$$\checkmark d(P_5, P_6) = \sqrt{(5-8)^2 + (6-1)^2} \approx 5.83.$$

→ closest pair : $(1, 2)$ & $(2, 3) \Rightarrow \sqrt{2} \approx 1.41.$

$(6, 7)$ & $(5, 6) \Rightarrow \sqrt{2} \approx 1.41.$

$$\rightarrow n=6 \quad \frac{n(n-1)}{2} = \frac{6 \times 5}{2} = 15 \text{ comparisons.}$$

Time complexity : $O(n^2).$

(76). CONVEX HULL.

PROBLEM INTERPRETATION & CONCEPT OF CONVEX BOUNDARY.

→ Given set of cartesian coordinate S :
 $\{(0, 0), (2, 5), (5, 7), (8, 4), (7, 0), (4, 2), (3, 3)\}$ we need
 to find subset of points that form convex
 hull.

→ Let test all pairs of directed lines (A,B) to find hull edges.

→ $A(0,0)$, $B(2,5)$, $C(5,7)$, $D(8,4)$, $E(7,0)$, $F(4,2)$, $G(3,3)$.

✓ Line $(0,0) \rightarrow (2,5)$.

✓ Line $(2,5) \rightarrow (5,7)$.

✓ Line $(5,7) \rightarrow (8,4)$.

✓ Line $(8,4) \rightarrow (7,0)$.

✓ Line $(7,0) \rightarrow (0,0)$.

→ Convex hull vertices: $(0,0) \rightarrow (2,5) \rightarrow (5,7) \rightarrow (8,4) \rightarrow (7,0) \rightarrow (0,0)$.

→ Choosing 2 points out of $n=7$ gives $\binom{7}{2} = 21$ possible undirected edges.

→ We test remaining $n-2=5$ points.

Total $42 \times 5 = 210$ individual tests.

Time complexity: $\mathcal{O}(n^3)$.

Q75. Closest Pair – Hospital Location Planning

1. Problem Interpretation and Justification of Brute Force Approach

- **Problem Interpretation:** We are given a set of hospital coordinates in a city map: $P = \{(1,2), (3,5), (6,7), (2,3), (8,1), (5,6)\}$. Our goal is to find the pair of hospitals that have the minimum Euclidean distance between them.
- **Justification of Brute Force:** The brute force approach calculates the distance between every possible pair of points in the dataset and selects the pair with the minimum distance. While more efficient algorithms like Divide and Conquer exist ($O(n \log n)$), the brute force method is straightforward, easy to implement, and ideal for very small datasets ($n = 6$) where overhead of complex algorithms is unnecessary.

2. Distance Computations Between All Possible Pairs

Using the Euclidean distance formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Let the points be labeled as:

- $P_1 = (1,2)$
- $P_2 = (3,5)$
- $P_3 = (6,7)$
- $P_4 = (2,3)$
- $P_5 = (8,1)$
- $P_6 = (5,6)$

Calculations for all unique pairs:

- $d(P_1, P_2) = \sqrt{(3-1)^2 + (5-2)^2} = \sqrt{4+9} = \sqrt{13} \approx 3.61$
- $d(P_1, P_3) = \sqrt{(6-1)^2 + (7-2)^2} = \sqrt{25+25} = \sqrt{50} \approx 7.07$
- $d(P_1, P_4) = \sqrt{(2-1)^2 + (3-2)^2} = \sqrt{1+1} = \sqrt{2} \approx 1.41$
- $d(P_1, P_5) = \sqrt{(8-1)^2 + (1-2)^2} = \sqrt{49+1} = \sqrt{50} \approx 7.07$
- $d(P_1, P_6) = \sqrt{(5-1)^2 + (6-2)^2} = \sqrt{16+16} = \sqrt{32} \approx 5.66$
- $d(P_2, P_3) = \sqrt{(6-3)^2 + (7-5)^2} = \sqrt{9+4} = \sqrt{13} \approx 3.61$
- $d(P_2, P_4) = \sqrt{(2-3)^2 + (3-5)^2} = \sqrt{1+4} = \sqrt{5} \approx 2.24$
- $d(P_2, P_5) = \sqrt{(8-3)^2 + (1-5)^2} = \sqrt{25+16} = \sqrt{41} \approx 6.40$
- $d(P_2, P_6) = \sqrt{(5-3)^2 + (6-5)^2} = \sqrt{4+1} = \sqrt{5} \approx 2.24$
- $d(P_3, P_4) = \sqrt{(2-6)^2 + (3-7)^2} = \sqrt{16+16} = \sqrt{32} \approx 5.66$
- $d(P_3, P_5) = \sqrt{(8-6)^2 + (1-7)^2} = \sqrt{4+36} = \sqrt{40} \approx 6.32$
- $d(P_3, P_6) = \sqrt{(5-6)^2 + (6-7)^2} = \sqrt{1+1} = \sqrt{2} \approx 1.41$

- $d(P_4, P_5) = \sqrt{(8 - 2)^2 + (1 - 3)^2} = \sqrt{36 + 4} = \sqrt{40} \approx 6.32$
- $d(P_4, P_6) = \sqrt{(5 - 2)^2 + (6 - 3)^2} = \sqrt{9 + 9} = \sqrt{18} \approx 4.24$
- $d(P_5, P_6) = \sqrt{(5 - 8)^2 + (6 - 1)^2} = \sqrt{9 + 25} = \sqrt{34} \approx 5.83$

3. Identification of Closest Pair and Comparison Count

- **Closest Pair:**

- Pair 1: (1,2) and (2,3) with distance $\sqrt{2} \approx 1.41$
- Pair 2: (6,7) and (5,6) with distance $\sqrt{2} \approx 1.41$

- **Total Comparisons Performed:** For $n = 6$ points, the number of pairs evaluated is given by $\frac{n(n-1)}{2} = \frac{6 \times 5}{2} = 15$ comparisons.

4. Time Complexity Analysis and Significance

- **Time Complexity:** The brute force method evaluates every unique pair, yielding a time complexity of $O(n^2)$.
- **Significance in Healthcare Access:** Identifying the closest pair of hospitals helps healthcare administrators find clusters or redundancies in medical resource distribution. High proximity can indicate overlapping service areas, whereas identifying isolated pairs ensures balanced emergency response coverage across the city.

Q76. Convex Hull – Boundary Detection in Urban Layout

1. Problem Interpretation and Concept of Convex Boundary

- **Problem Interpretation:** Given a set of urban coordinate points $S = \{(0,0), (2,5), (5,7), (8,4), (7,0), (4,2), (3,3)\}$, we need to find the subset of points that form the convex hull (the outer boundary enclosing all points).
- **Concept:** The convex hull of a set of points is the smallest convex polygon that contains all the points. In a brute force approach for a 2D plane, an edge formed by any two points belongs to the convex hull if and only if all other points in the set lie on or to one side of the directed line passing through the two points.

2. Edge Testing and Reasoning

To find the hull edges, we test all possible pairs of directed lines (A, B) to check if all remaining points lie entirely on one side (i.e., have the same non-negative or non-positive cross-product orientation).

Let the points be:

- $A(0,0), B(2,5), C(5,7), D(8,4), E(7,0), F(4,2), G(3,3)$

Evaluating potential boundary edges using orientation tests:

- **Line $(0,0) \rightarrow (2,5)$:** Other points check out. This forms an edge.
- **Line $(2,5) \rightarrow (5,7)$:** Other points lie on one side. This forms an edge.

- **Line** $(5,7) \rightarrow (8,4)$: Other points lie on one side. This forms an edge.
- **Line** $(8,4) \rightarrow (7,0)$: Other points lie on one side. This forms an edge.
- **Line** $(7,0) \rightarrow (0,0)$: Other points lie on one side. This forms an edge.

The internal point $F(4,2)$ and $G(3,3)$ are successfully distinguished because lines passing through them or tested against them will have other points falling on both left and right sides, failing the single-side condition.

- **Convex Hull Vertices (in counter-clockwise order):**
 $(0,0) \rightarrow (2,5) \rightarrow (5,7) \rightarrow (8,4) \rightarrow (7,0) \rightarrow (0,0)$

3. Comparison Count and Time Complexity Analysis

- **Total Comparisons:**
 - Choosing 2 points out of $n = 7$ gives $\binom{7}{2} = 21$ possible undirected edges (or 42 directed edges).
 - For each directed edge, we test the remaining $n - 2 = 5$ points.
 - Total point-position checks = $42 \times 5 = 210$ individual orientation evaluations.
- **Time Complexity:** The brute force convex hull algorithm has a time complexity of $O(n^3)$ because it checks every pair of points ($O(n^2)$) against all other points ($O(n)$).

4. Practical Significance in Urban Mapping

- **Urban Mapping Significance:** Convex hull algorithms are crucial for determining city limits, municipal

boundaries, zoning districts, or fencing geographic service areas. It simplifies complex spatial datasets by enclosing outer limits, helping urban planners outline boundaries efficiently.
